

Unit I

COMPONENTS OF AN OS

Course Objective – Cob1: To understand the basics and functions of operating systems.

Course Outcome - CO1 : Classify operating system components and system calls based on functionality

Part A

S.No.	Question	BTL *
1.	List the four main elements of a computer. • Processor – Controls the operation of the computer & performs its data processing functions • Main memory – Stores data & programs. it is volatile. • I/O modules – Move data between the computer & its external environment such as disks, communication equipment & terminals. • System Bus – Provides for communication among processors, main memory & I/O modules.	K1
2	Recall operating system. An operating system is a program that manages the computer hardware. It acts as an intermediate between a user's of a computer and the computer hardware. It controls and coordinates the use of the hardware among the various application programs for the various users.	K1
3.	Name any three main objectives of operating system design. • Convenience – An OS makes a computer more convenient to use • Efficiency – An OS allows the system resources to be used in efficient manner • Ability to Evolve – An OS Constructed in such a way as to permit the effective development, testing & introducing new function.	K1
4.	List the services provided by an operating system. 1. Program development 2. Program execution 3. User Interface 4. I/O Operations 5. File system Manipulation 6. Communication 7. Error Detection 8. Resource allocation 9. Accounting 10. Security	K1

5.	Recall the various modes of an operating system. The two modes of OS are user mode and kernel mode. The user mode helps the operating system in running user applications. The kernel model is required when the system boots and operating system is loaded. Some of the privileged instructions work in kernel mode only.	K1												
6.	State the definition of an Interrupt. Interrupt are provided primarily as way to improve processor utilization. It is a mechanism by which other modules(I/O, Memory) may interrupt the normal sequencing of the processor. EG: moving a mouse or pressing a keyboard key. Classes of interrupts: • Program • Timer • I/O • Hardware failure	K1												
7.	Illustrate the concept of a trap in operating systems with an example. In computing and operating systems, a trap, also known as an exception or a fault is typically a type of synchronous interrupt caused by an exceptional condition (e.g., breakpoint, division by zero, invalid memory access).	K2												
8.	List the advantages of Multiprocessor Systems. The advantages of the multiprocessing system are: • Increased Throughput – By increasing the number of processors, more work can be completed in a unit time. • Cost Saving – Parallel system shares the memory, buses, peripherals etc. Multiprocessor system thus saves money as compared to multiple single systems	K1												
9.	Differentiate between Multiprogramming and Multitasking. <table border="1"><thead><tr><th>Multiprogramming</th><th>Multitasking</th></tr></thead><tbody><tr><td>Uses a single CPU to execute programs.</td><td>Uses multiple tasks for task allocation.</td></tr><tr><td>Concept of Context Switching is used.</td><td>Concepts of Context Switching and Time Sharing are used.</td></tr><tr><td>Increases CPU utilization by organizing jobs.</td><td>Increases CPU utilization and also increases responsiveness.</td></tr><tr><td>The goal is to reduce CPU idle time as long as possible.</td><td>The goal is also to reduce CPU idle time as long as possible.</td></tr><tr><td>Execution of a process takes more time.</td><td>Execution of a process takes less time.</td></tr></tbody></table>	Multiprogramming	Multitasking	Uses a single CPU to execute programs.	Uses multiple tasks for task allocation.	Concept of Context Switching is used.	Concepts of Context Switching and Time Sharing are used.	Increases CPU utilization by organizing jobs .	Increases CPU utilization and also increases responsiveness .	The goal is to reduce CPU idle time as long as possible.	The goal is also to reduce CPU idle time as long as possible.	Execution of a process takes more time .	Execution of a process takes less time .	K2
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10.	State System Call and list their categories. System Call provides the interface between running program and the OS User can request any services from OS through System Call. List the categories of system call are: • File management • Process Management • Inter process Communication • I/O Device Management • Information Processing & Maintenance	K1						
11.	State System Programs and list their categories. System programs provide a convenient environment to the user for developing and executing the programs. Categories: - 1. File management 2. Status Information 3. File Modification 4. Programming language support 5. Program loading & execution Communication	K1						
12.	List the advantages of Multiprogramming. Multi Programming increases CPU Utilization by organizing jobs so that the CPU always has one to execute. Advantage: • It increases CPU utilization. • It makes efficient use of the CPU overlapping the demands for the CPU & I/O devices. • Increased throughput and Lower response time.	K1						
13.	Recall Bootstrapping. The boot program is stored on disk with predetermined address called boot sector. The boot program then loads the operating system into memory to startup the computer, this arrangement is known as bootstrapping.	K1						
14.	Why do programmers typically use APIs rather than invoking system calls directly? System calls are much slower than APIs (library calls) since for each system call, a context switch has to occur to load the OS (which then serves the system call). Most details of OS interface hidden from programmer by API Managed by run-time support library (Set of functions built into libraries included with compiler.)	K2						
15.	Give an example of a system program used for file modification. A text editor, such as vi, nano, or Notepad, is a common system program used for creating and modifying the contents of text files.	K1						
16.	Differentiate between Batch systems and Time-sharing systems. <table><tr><th>Batch System</th><th>Time Sharing System</th></tr><tr><td>Jobs or work are kept in order and run one after the other (sequentially).</td><td>Tasks are given specific time slices, and the OS switches between different tasks.</td></tr><tr><td>No user interaction during processing.</td><td>User interaction is involved during processing.</td></tr></table>	Batch System	Time Sharing System	Jobs or work are kept in order and run one after the other (sequentially).	Tasks are given specific time slices , and the OS switches between different tasks.	No user interaction during processing.	User interaction is involved during processing.	K2
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17.	List the types of OS Structure. ->Simple Structure ->Kernel Structure • Monolithic • Microkernel • Hybrid Kernel • Exo-Kernel ->Layered Structure	K1																					
18.	Compare a Micro kernel with a Monolithic kernel. <table border="1"> <thead> <tr> <th>Basis for Comparison</th><th>Micro kernel</th><th>Monolithic Kernel</th></tr> </thead> <tbody> <tr> <td>Size</td><td>Microkernel is smaller in size</td><td>It is larger than microkernel</td></tr> <tr> <td>Execution</td><td>Slow execution</td><td>Fast execution</td></tr> <tr> <td>Extensibility</td><td>It is easily extendible</td><td>It is hard to extend</td></tr> <tr> <td>Security</td><td>If a service crashes, it affects only the microkernel</td><td>If a service crashes, the whole system crashes</td></tr> <tr> <td>Code Complexity</td><td>More code is required to write a microkernel</td><td>Less code is required to write a monolithic kernel</td></tr> <tr> <td>Examples</td><td>QNX, Symbian, L4Linux etc.</td><td>Linux, BSDs (FreeBSD, OpenBSD, NetBSD) etc.</td></tr> </tbody> </table>	Basis for Comparison	Micro kernel	Monolithic Kernel	Size	Microkernel is smaller in size	It is larger than microkernel	Execution	Slow execution	Fast execution	Extensibility	It is easily extendible	It is hard to extend	Security	If a service crashes, it affects only the microkernel	If a service crashes, the whole system crashes	Code Complexity	More code is required to write a microkernel	Less code is required to write a monolithic kernel	Examples	QNX, Symbian, L4Linux etc.	Linux, BSDs (FreeBSD, OpenBSD, NetBSD) etc.	K2
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19.	How does interrupt differ from trap? • An interrupt is a hardware-generated signal that changes the flow within the system. A trap is a software-generated interrupt. • An interrupt can be used to signal the completion of I/O so that the CPU doesn't have to spend cycles polling the device. A trap can be used to catch arithmetic errors or to call system routines	K2																					
20.	Why does time-sharing differ from multiprogramming? Main difference between multi programming and time sharing is that multiprogramming is the effective utilization of CPU time; by allowing several programs to use the CPU at the same time but time sharing is the sharing of a computing facility by several users that want to use the same facility at the same time.	K2																					

Part B

S.No.	Question	BTL *
1.	Explain different types of Operating System Structures. (16)	K2
2.	Outline the system calls. Explain how system calls help user programs to interact with the OS. (16)	K2
3.	Explain the evolution of Operating Systems.(8)	K2
4.	Describe the overview of a Computer System. (8)	K2
5.	Describe the key services provided by an Operating System. (8)	K2
6.	Explain the categories of Operating System Programs. (8)	K2

7.	Explain the role of an Operating System as a resource manager. Discuss how the OS manages CPU, memory, and I/O devices in a computer system. (16)	K3
8.	Case Study: The "Quick Kernel" Startup Initiative A tech startup, "Quick Kernel," is developing a new, lightweight operating system for a novel line of budget educational tablets. The tablets have limited hardware resources (single-core processor, 2GB RAM). The OS must be efficient, secure, and user-friendly for students. The development team is in the initial design phase and is debating fundamental architectural choices. One faction advocates for a simple, monolithic structure for maximum performance and easier initial development. Another faction pushes for a microkernel structure, arguing it will make the OS more stable, secure, and easier to update with new services (like a parental control module) in the future. Furthermore, the team is designing the interface for educational app developers. They need to decide how these third-party apps will request essential services from the OS—such as writing to a file, drawing on the screen, or getting network data—without being allowed direct, unprotected access to the hardware. Based on the case study, answer the following: - Explain the monolithic and microkernel operating system structures. Compare their advantages and disadvantages in the context of Quick Kernel's budget tablet constraints and long-term goals (like stability and updatability). (8) - Outline the concept of system calls. Explain the step-by-step process of how a student's drawing app, using a <code>write()</code> system call, would interact with the Quick Kernel OS to save a file. Your explanation should clearly show the transition from user mode to kernel mode and back, and how this mechanism protects the system. (8)	K2